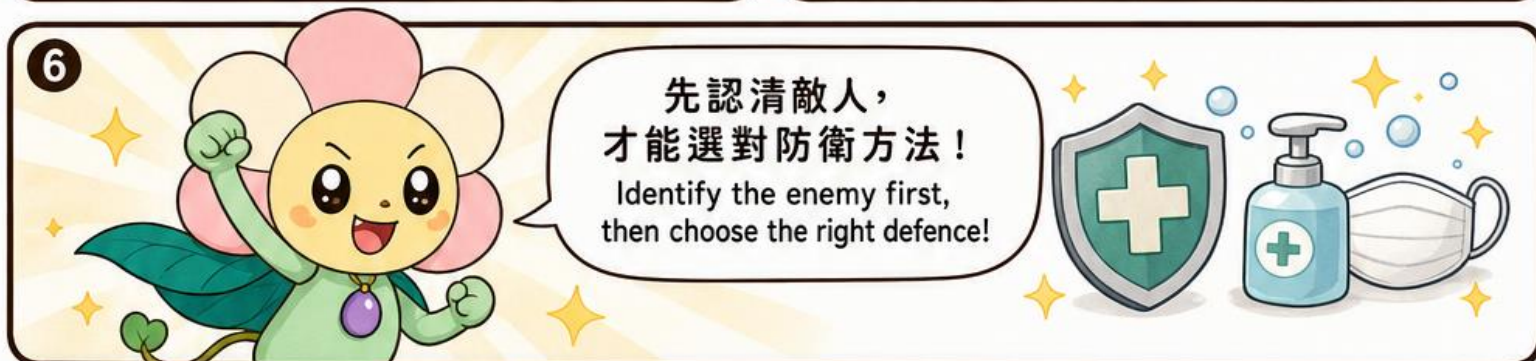
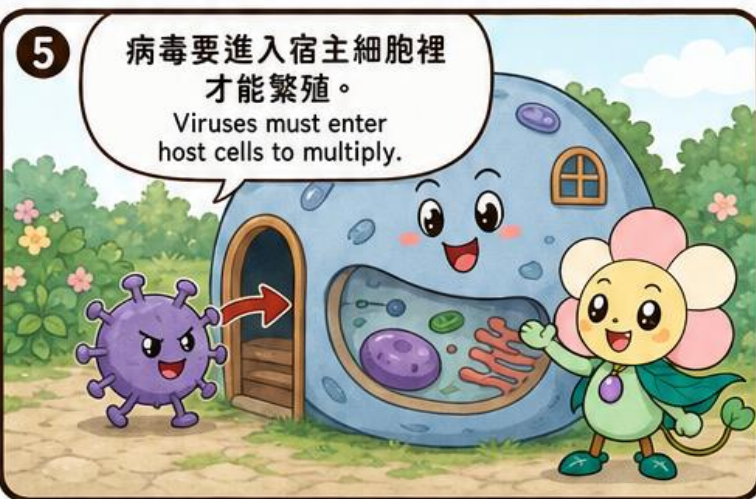
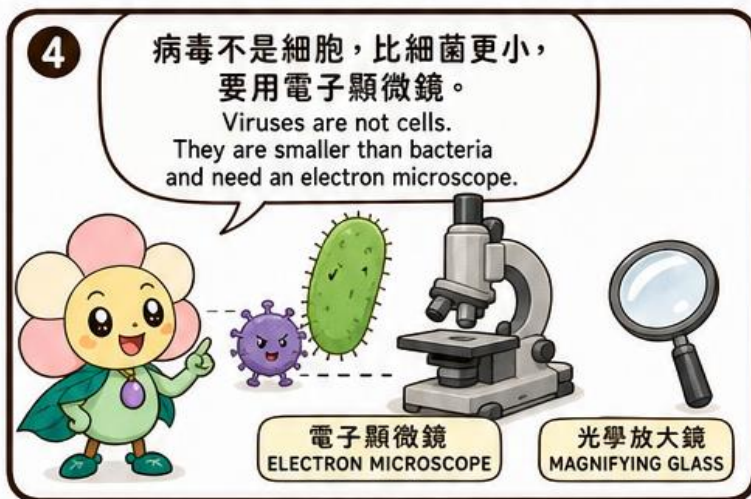
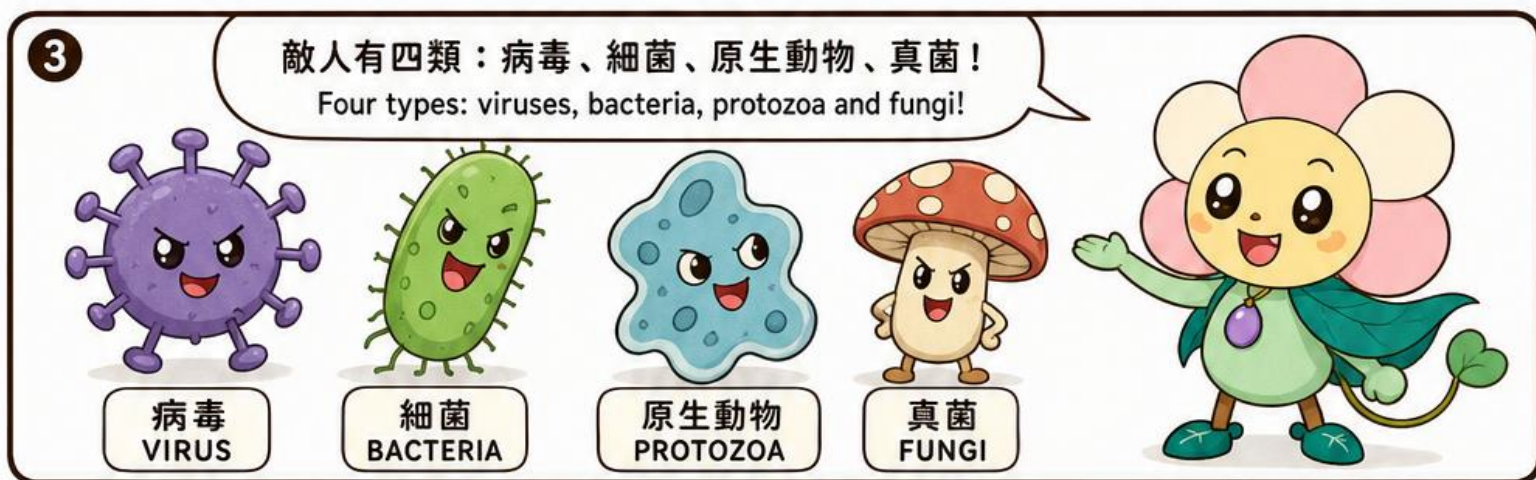


第1頁 PAGE 1 — 病原體警報 / THE PATHOGEN ALARM

花瓣仔與病原體防衛任務 / Petalino and the Pathogen Defence Mission



傳染病 = 病原體引起的疾病 / Infectious diseases are caused by pathogens

這個病毒走體液路線！
This virus travels
the body-fluid route!

體液
BODY FLUIDS

HIV 可引起愛滋病，經體液、
性接觸和母嬰傳播。
HIV can cause AIDS. It spreads by
body fluids, sexual contact, and
mother-to-child transmission.



體液
Body fluids



性接觸
Sexual
contact



母嬰傳播
Mother-to-child
transmission

預防：採取安全性行為，
例如使用安全套。
Prevention: safer sexual
practices, for example
using a condom.

安全性行為
更安心！
SAFER SEX
BETTER CHOICES
BETTER HEALTH

安全套
CONDOM

不要共用針具！
也要保護開放性傷口！
Do not share needles!
Protect open wounds too!



乙型肝炎也經體液傳播。
不要共用個人物品！
Hepatitis B also spreads through
body fluids. Do not share
personal items!

乙型肝炎
Hepatitis B

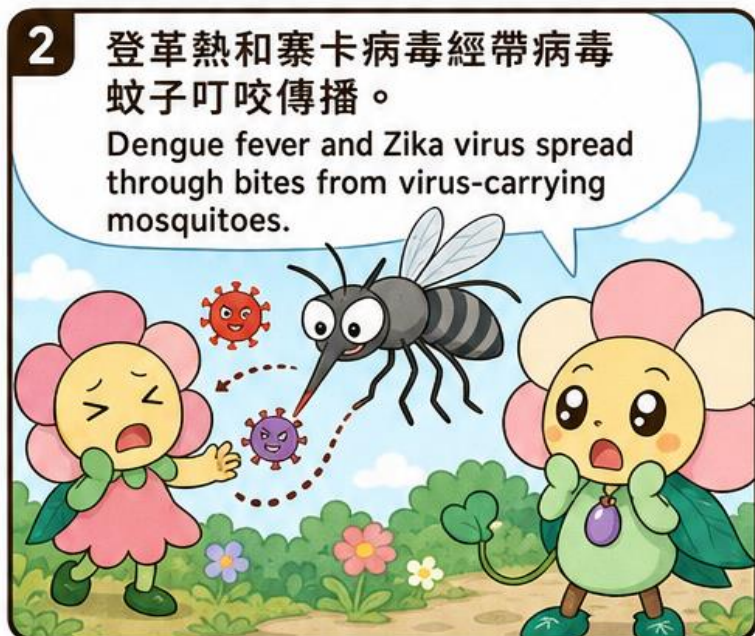


乙型肝炎可以接種疫苗來預防！
Hepatitis B can be prevented
by vaccination!

接種疫苗
保護自己
保護他人！
VACCINATION
PROTECTS YOU
PROTECTS OTHERS!



體液傳播 → 保護傷口、針具和個人物品
Body-fluid route → protect wounds, needles and personal items

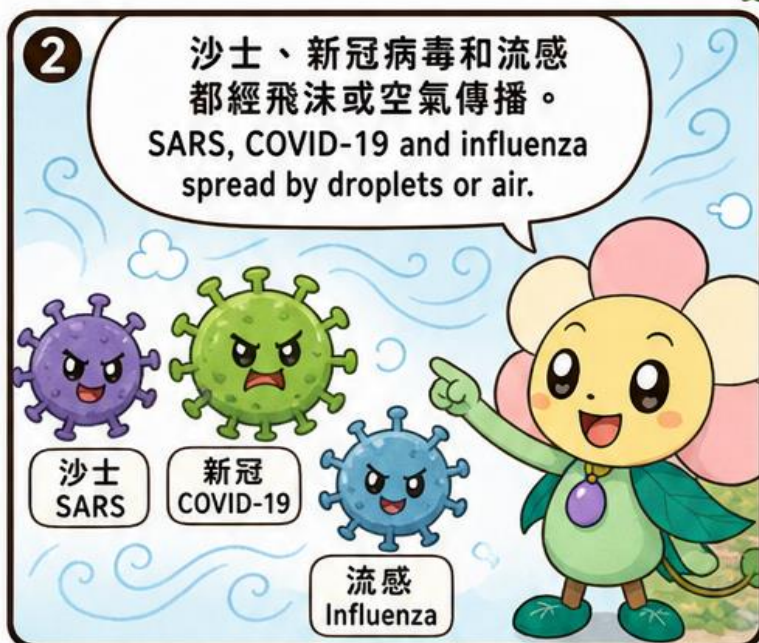


蚊子叮咬
Mosquito bites



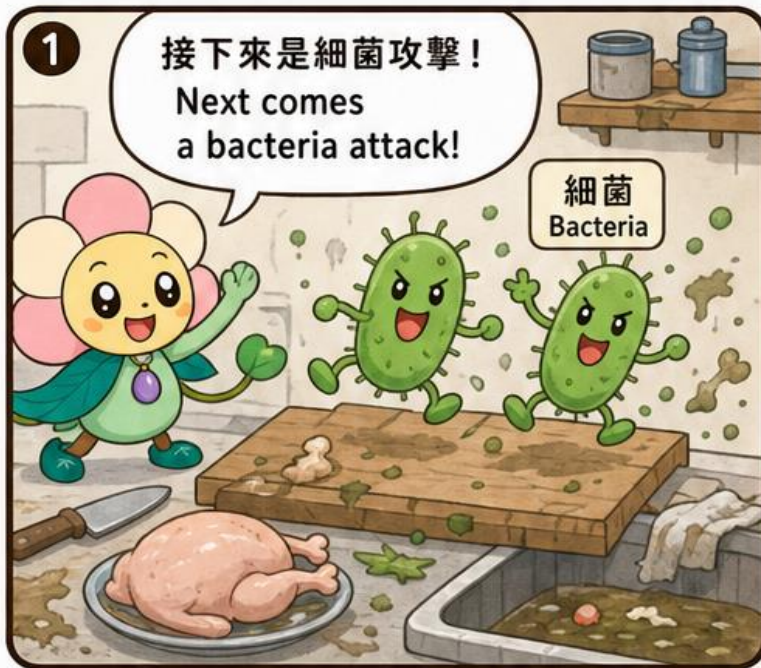
防蚊措施
mosquito prevention





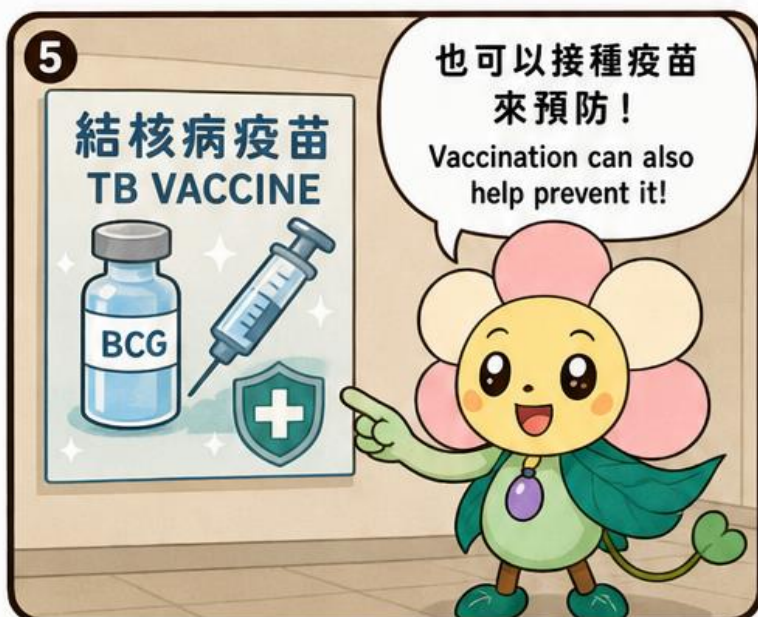
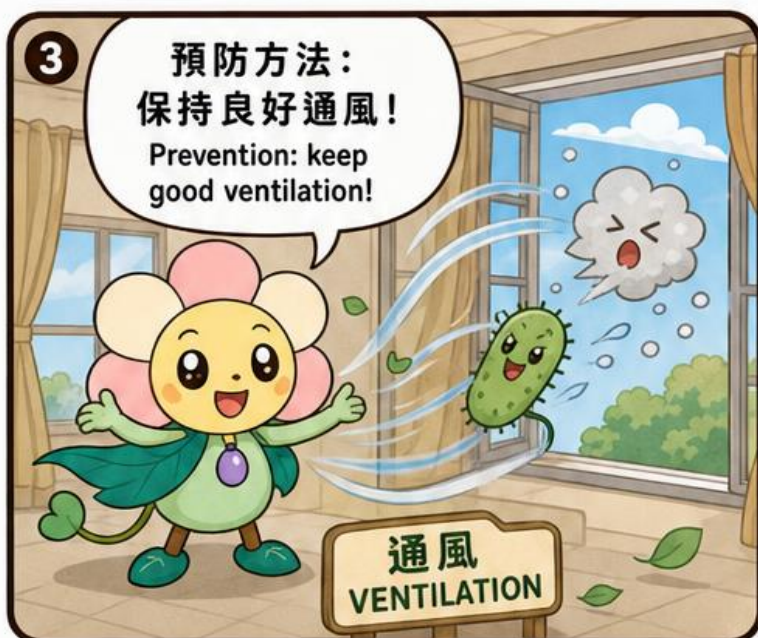
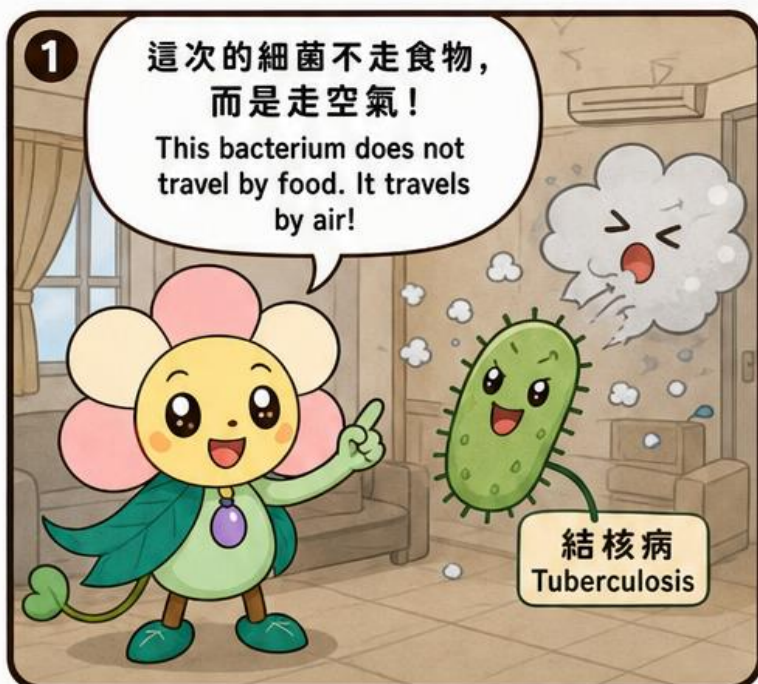
飛沫、空氣、接觸 → 口罩、洗手、疫苗、保持距離

Droplets, air, contact → mask, handwashing, vaccine, distance

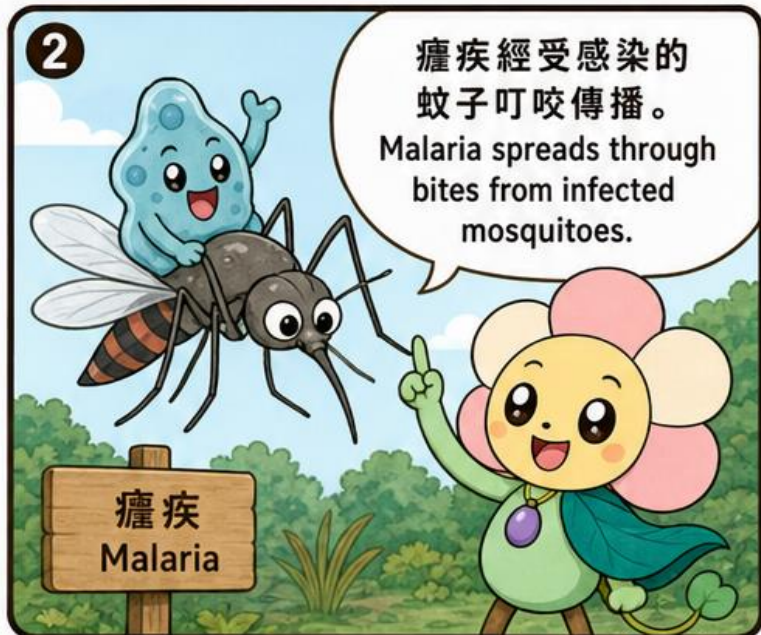


受污染食水 → 衛生、漁淨食水、徹底煮熟 /
Contaminated food and water → hygiene, clean supplies, thorough cooking





結核病 → 空氣飛沫 → 疫苗、通風、口罩 /
TB → airborne droplets → vaccine, ventilation, mask



瘧疾 → 蚊帳驅蚊劑 / 阿米巴痢疾 → 安全食水與食物衛生
Malaria → nets and repellent / Dysentery → safe water and food hygiene



腳癬 → 直接接觸／污染物件 → 乾爽、透氣、不共用、不赤腳
Athlete's foot → contact or objects → dry, breathable, no sharing, no barefoot

1

爆發控制住了！
現在來學習疫苗怎樣運作。
The outbreak is under control!
Now let us learn how vaccines work.



2

疫苗預先建立免疫記憶，
真正病原體來時就能迅速反應。
Vaccines create immune memory
in advance, so the body can
respond rapidly later.



3

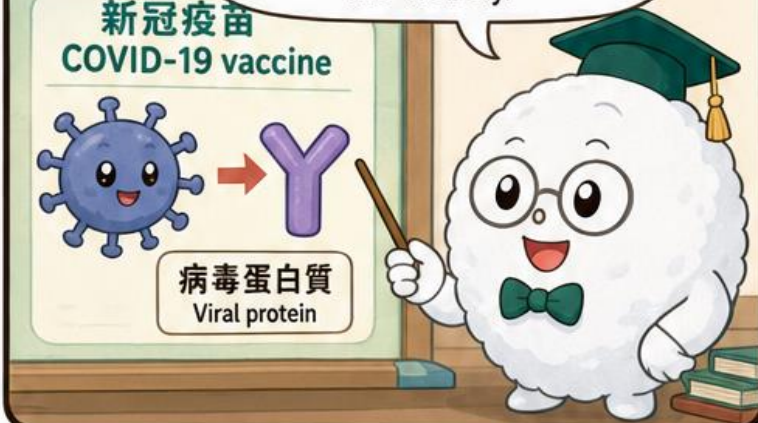
步驟1 STEP 1

第一步：接種疫苗。
疫苗含有病原體片段或已滅活的病原體。
Step 1: the vaccine is given.
It contains fragments or an
inactivated pathogen.



4

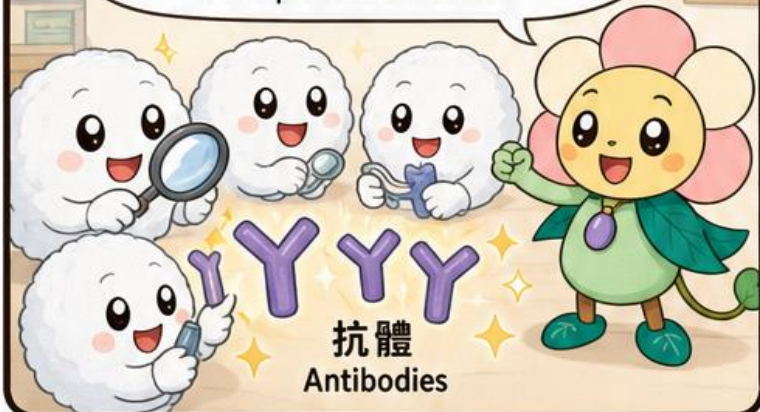
例如新冠疫苗，
把病毒的蛋白質引入身體。
For example, the COVID-19 vaccine
introduces a protein of the virus
into the body.



5

步驟2 STEP 2

第二步：白血球辨認後製造抗體。
Step 2: white blood cells recognise
it and produce antibodies.



6

步驟3 STEP 3

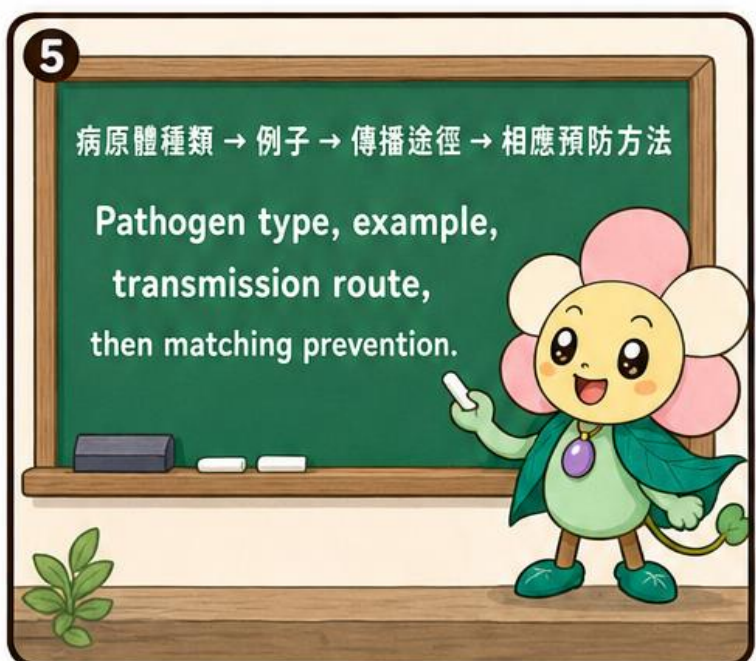
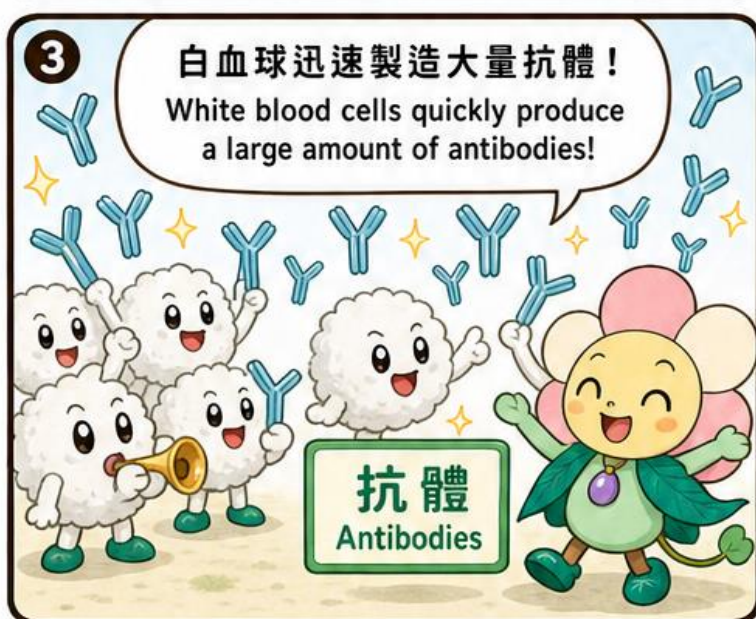
第三步：形成記憶細胞，
記住這個病原體！
Step 3: memory cells form
and remember this pathogen!



疫苗 = 安全綵排 → 抗體 + 記憶細胞

A vaccine is a safe rehearsal → antibodies + memory cells





接種後再遇病原體 → 快速製造大量抗體
After vaccination, later infection meets a rapid antibody response